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EDUCATION

- **Master in AIS – Artificial Intelligence Systems** 2025–2027
EPITA, Paris, France
- **Bachelor of Technology in Artificial Intelligence and Machine Learning** 2021–2025
Kalasalingam Academy of Research and Education, TamilNadhu, India

PERSONAL PROJECTS

- **Renewable Energy Consumption on Solar & Wind Energy Prediction**
Deep learning and Generative AI to forecast solar and wind energy using NASA POWER dataset.
 - Built deep learning models to predict solar and wind energy using NASA POWER dataset (2014–2024).
 - Implemented LSTM and ARIMA models to improve prediction accuracy using feature engineering and time-series preprocessing.
 - Technology Used: Python, TensorFlow, LSTM, ARIMA, Generative AI.
- **Bike Demand Forecasting**
Full ML pipeline with PostgreSQL-backed dataset for time-series demand prediction.
 - Designed a full ML pipeline with PostgreSQL-backed dataset for time-series demand prediction.
 - Implemented SVM and Random Forest models and deployed an interactive dashboard using Streamlit for real-time predictions.
 - Technology Used: Python, SVM, Random Forest, PostgreSQL, Streamlit.
- **RAG-Powered Multi-Agent QA System with MCP Server**
Two-phase LLM system combining corpus-grounded RAG with LLM-as-judge evaluation and 3 autonomous AI agents.
 - Phase 1: Fine-tuned an LLM on a custom corpus using RAG; model retrieves and answers strictly from corpus data, with a second LLM acting as an evaluator/judge to verify response accuracy and relevance.
 - Phase 2: Designed and deployed 3 autonomous AI agents connected via an MCP server – agents collaboratively train the model, retrieve information from the corpus, and perform live web search with cited sources.
 - Technology Used: Python, LangChain, RAG, LLM Fine-tuning, MCP, AI Agents, Vector Database.
- **Automated Pneumonia Detection from Chest X-Rays Using Deep Learning**
Transfer learning-based medical image classification with Flask web app and Docker deployment.
 - Implemented transfer learning (ResNet50/EfficientNet) for multi-class X-ray classification achieving 94% accuracy and 0.97 AUC.
 - Built a Flask web app with Grad-CAM explainability and Docker deployment for real-time predictions.
 - Technology Used: Python, PyTorch, ResNet50, EfficientNet, Flask, Docker.

EXPERIENCE

- **Machine Learning Internship** 2023
Teachnook / Croatia Tech Online
 - Developed a ML Model with 95% accuracy and deployed it via a Flask web app.
 - Optimized model performance with data preprocessing and collaborated on real-world validation.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, Java, SQL

Libraries: TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy

Web Dev Tools: Flask, Streamlit, Docker, Git, GitHub

Frameworks: Apache Airflow, Gymnasium

Cloud/Databases: PostgreSQL, MySQL

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Time-Series Analysis, Reinforcement Learning, Data Science for Production.

Areas of Interest: Machine Learning, Deep Learning, Renewable Energy Forecasting, Computer Vision, Generative AI, LLMs.

Soft Skills: Problem Solving, Team Lead, Collaboration, Analytical Thinking

PUBLICATIONS

- **Renewable Energy Consumption on Solar and Wind Energy Prediction** (Link) IEEE, 2024
 - Published IEEE research on forecasting solar and wind energy output using deep learning and generative AI techniques.